STALLINGS FOLDINGS FOR SUBMONOIDS OF AUTOMATIC GROUPS

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Abstract. [insert actual abstract] The authors would like to thank Marco Linton and Armando Martino for their useful input in the application to surface groups. The authors acknowledge support by Leverhulme foundation grant no.RPG-2022-025 during the work on this article.

1. Introduction

This paper describes a procedure whose aim is to construct an automaton to recognise the elements of a specified submonoid or rational subset of an automatic group, using techniques similar to those of Stallings' folding algorithm, which applies to subgroups of free groups andd their generalisations. More precisely we develop the partial algorithm of [11, Section 4], which applies to a subgroup H of an automatic group G with associated regular language L, and terminates with an automaton recognising H precisely when H is L-quasi-convex; note that, since it produces the required output whenever it terminates, this procedure satisfies the criteria to be a partial algorithm. The concept of L-quasi-convexity (which we often abbreviate as L-qc) is defined in [11], generalising the concept of quasi-convexity; when G is hyperbolic and L is the set of all geodesics, L-quasi-convexity is simply quasi-convexity.

SER: tried to add criteria for partial algorithm

Given an automatic group G, finitely generated over S, and with associated regular language L of words over S mapping onto G, we define a procedure that extends the partial algorithm of [11] to solve the membership problem for a finitely generated subgroup of G to the setting of a rational subset K of G that might not be a subgroup. Our generalisation requires a condition on K that is stronger than L-quasi-convexity, which we call *L-proximacy* (defined in Section 3) where, as in [11], L is a regular subset of S^* mapping naturally onto G. For an L-proximate rational subset K of G we construct a procedure which will, after a finite number of steps, output an automaton accepting the language of all words in L that map to K. In the case where K is an L-proximate submonoid of G (necessarily also rational), we are able to turn this procedure into an algorithm, since we can test, at each iteration of the procedure, to see whether or not it has completed. In addition, for G in various classes of groups (RAAGs, surface groups) we are also able to make our procedure into an algorithm for L-proximate submonoids of G, and we describe how to do this.

In Section 2 we make preliminary definitions and give some background on automata, rational subsets of groups, and convexity. In Section 3 we introduce the notion of an L-proximate subset of a group and establish the basic properties needed for our purposes. In Section 4 we describe our procedure to construct an automaton recognising the L-representatives of a rational subset of a group G. In Section 5 we discuss how and when it is possible to determine when our procedure has completed, giving some particular examples which we believe are not covered in the literature, for example of submonoids of surface groups with solvable membership problem.

1.1. **Background: Stallings folding.** The concepts of Stallings foldings and Stallings automata, when introduced in 1983 [14], provided group theorists with a completely new approach to understanding finitely generated subgroups of free groups.

Given a free group G with generating set S, and a subgroup $H = \langle T \rangle$ of G, specified by its finite generating set T of words over S, Stallings' algorithm applies a sequence of 'foldings' to a labelled graph constructed from T, iterating until it yields a deterministic finite state automaton (or labelled graph) $\mathrm{St}(H)$, called the *Stallings automaton* of H, which recognises the set of all words over S that are within the subgroup H. From an automata theoretic point of view, the folding operations are precisely what is needed for determinisation and minimisation of an finite state automaton with the property that the reverse of a transition on a symbol x is a transition on the symbol x^{-1}; following [1] we shall call such an automaton *involutive*.

From a graph theoretic point of view, $\mathrm{St}(H)$ is the core of the Schreier graph of H.

The Stallings automaton for a subgroup provides more than just a finite object to recognise its elements; a lot of valuable information can be read from it. Information about the subgroup index, conjugates, normaliser, decidability problems regarding membership and intersections, separability and much more can be easily extracted from it.

The overall framework of a generalisation of Stallings folding to subgroups of groups beyond free groups is to have a group G with finite generating set S and a generating set T of words over S for a subgroup $H \leq G$, and to then iteratively perform an operation (folding) on a structure built from T until a finite object $\mathrm{St}(H)$ is reached, that can be uniquely associated to H, from which we can extract information about H. The subgroup H might need to satisfy certain hypotheses to guarantee that this finite object is reached; in the case of free groups this amounts to asking for H to be finitely generated. Given how successful this approach was in the context of free groups, it comes as no surprise that generalisations have been developed to try to understand subgroups of other families of groups.

Recently, Dani–Levicovitz [5] used similar techniques to study subgroups of right-angled Coxeter groups, using a generalisation of folding that yields a Stallings-like object which is a cube complex instead of a graph. Information about normality and index is reflected in this cube complex and, in the case that the subgroup is finitely generated and quasi-convex, the associated cube complex is finite and this information can be explicitly read from that finite object. Generalising this, Ben-Zvi–Kropholler–Lyman [3] developed a Stallings foldings framework to study subgroups of fundamental groups of non-positively curved cube complexes. This construction also yields a finite object when the subgroup is finitely generated and quasi-convex, and this object too encodes information about normality and index, and allows solution of the subgroup membership problem. Further away from cube complexes, the work by Kharlampovich–Miasnikov–Weil [11] gives another generalisation of Stallings foldings for automatic groups, solving in this way the subgroup membership problem for H and other questions regarding finiteness of H, intersections of subgroups, etc.

2. Definitions and notation

Words, free monoids, free groups: **AD:** Given a set Σ we denote by Σ^* the set of all words on Σ, as well as the free monoid over Σ if we want to stress the concatenation operation. A set Σ equipped with a map ν to a monoid M is called a *monoid generating set* for M if the natural extension of ν to a map from Σ^* to M is surjective. Usually all mention of ν is suppressed and often we implicitly assume that Σ is a subset of M and ν is the inclusion map. The definition of a *group generating set* for a group G is made the same way, using the free group on Σ instead of the free monoid on Σ.

Throughout this document, unless stated otherwise, $S = \{s_1, \dots s_r\}$ will denote a finite alphabet. We denote by S^* the set of words over S, as well as the free monoid over S if we want to stress the concatenation operation. The symbols ε and 1 are respectively used to refer to the empty word and to the identity element (in a monoid or a group).

We denote by $S^{\pm} = S \sqcup S^{-1}$ the alphabet given by the letters in S together with their (formal) inverses. The free group on S, denoted $F(S)$, is given by the words in $(S^{\pm})^*$ under the equivalence relation of free reduction, with concatenation as the group operation. For each $w \in (S^{\pm})^*$ we denote by $\overline{w}$ its freely reduced representative. The free group $F(S)$ can be identified with the set of freely reduced words in $(S^{\pm})^*$, therefore throughout this document we will often see the free group as a subset of the free monoid over $S^{\pm}$. However we note that although $F(S)$ embeds as a subset in $(S^{\pm})^*$ it does not embed as a submonoid.

For a word w in $(S^{\pm})^*$ we denote by $|w|$ its length, given by the number of letters in w, and we write $|w|_{F(S)}$ for the length $|\overline{w}|$ of its reduced representative $\overline{w}$.

For a group G with generating set S, we denote by $\mu\colon (S^{\pm})^* \to G$ and $\pi\colon F(S) \to G$ the natural projections from the free monoid and the free group, respectively, onto G. Note that μ factors through π.

$$\begin{tikzcd}
(S^{\pm})^* \arrow[r, "\overline{(-)}"] \arrow[dr, "\mu"'] & F(S) \arrow[d, "\pi"] \\
 & G = \langle S|R\rangle
\end{tikzcd}$$

Understanding $F(S)$ as a subset of $(S^{\pm})^*$ we can think of π as the restriction $\mu|_{F(S)}$, and therefore of $\pi^{-1}(G)$ as a subset of $(S^{\pm})^*$. To denote that two words w_1, w_2 in $(S^{\pm})^*$ or $F(S)$ have the same image under μ or π, we will write $w_1 =_G w_2$.

Given a set of words $T \subset (S^{\pm})^*$, we denote by $T^{\pm}$ the union of T with the set T^{-1} of its formal inverses; each element of the subgroup $\langle T\rangle$ can be represented as a product of elements of $T^{\pm}$. For w in $(T^{\pm})^*$ (representing an element of the subgroup $\langle T\rangle$), we define the length of w in G with respect to T to be

$$|w|_{G,T} = \min\{n\colon w =_G v,\ v = h_1 \dots h_n,\ h_i \in T^{\pm}\}.$$

Finite state automata: We refer to [9] for notation and standard results, noting that more detail can be found in the classic text [10]. The notation $\mathcal{A} = (\Sigma, V, E, V_F, v_0)$ will be used to describe a *finite state automaton* (FSA) where Σ is a finite set (the *alphabet*), V is the set of vertices or *states*, E is the set of Σ-labelled edges or *transitions* of the form $(v, s, w) \in V \times \Sigma \times V$, $v_0 \in V$ is the *initial state* and $V_F \subset V$ is the set of *accepting states* of the automaton. As our notation suggests, we shall often view $\mathcal{A}$ as an edge-labelled directed graph. We say that (v, s, w) is an *s-labelled* transition, with v as its *source* state and w as its *target* state. For an edge $e = (v, s, w)$, we will use the notation $l(e) = s$ for the labelling function. Unless stated otherwise, the word automaton will implicitly refer to one with finitely many states, throughout this text.

To each finite state automaton $\mathcal{A}$ we can assign the set $L(\mathcal{A}) \subset \Sigma^*$ of words *accepted* by the automaton, which are the words given by the concatenation of edge labels of paths in $\mathcal{A}$ that start in v_0 and end in one of the accepting states; we call $L(\mathcal{A})$ the *language* accepted by $\mathcal{A}$, and say that $\mathcal{A}$ *recognises* $L(\mathcal{A})$. We say that a subset L of Σ^* is *regular* if it is the language $L(\mathcal{A}_L)$ accepted by a finite state automaton $\mathcal{A}_L$. Where M is a monoid with finite generating set Σ, we call a subset K of M *rational* if it is the image of a regular subset Q of Σ^* under **AD:** the canonical monoid homomorphism $\mu : \Sigma^* \to M$. We note that this is one of a few equivalent definition of rational subsets of monoids, and that a subset of Σ^* is regular precisely when it is rational.

Since we want to use automata to define languages that are subsets of groups, the alphabets Σ of our automata will often be sets of the form $S^{\pm} = S \cup S^{-1}$, where S is the finite generating set of a related group.

It is well known that if a subset L of $(S^{\pm})^*$ is regular, its *free reduction* $\overline{L}$ is also regular (Benois' theorem, see [2, 1],).

We can consider automata with ε-transitions, that is, automata over the alphabet $\Sigma_\varepsilon = \Sigma \sqcup \{\varepsilon\}$.

We note that, given any automaton $\mathcal{A}_\varepsilon$ over the alphabet Σ_ε, an algorithm described in [9, Section 2.5.1] constructs an automaton $\mathcal{A}$ over Σ accepting the same language.

Similarly, given a *non-deterministic FSA* (one where s-labelled transitions (v, s, w), (v, s, w') with different target states $w \neq w'$ are allowed) an algorithm described in [9, Proposition 2.5.2] constructs a *deterministic FSA*

From the above it is clear that there are many different FSA accepting a given regular set. However, there is a unique one that is deterministic and has the least number of states, which we call the *minimal automaton*, and there is an algorithm to obtain it from a non-minimal one (Myhill–Nerode) [9, Theorem 2.5.4]. Note that the minimal automaton is *trim*, that is, that each state in the automaton is used to read at least one word accepted by the automaton (so there are no failure or unreachable states).

Rational structures and automatic groups: A pair (G, L) is called a *rational structure* for a group G if $G = \langle S \rangle$ for a finite alphabet S and $L \subset (S^{\pm})^*$ is a regular set where the restriction $\mu|_L : L \to G$ is a surjection. We note that if (G, L) is a rational structure for G, then $(G, \bar{L})$ is also a rational structure, with $\bar{L} \subset F(S)$.

Given a rational structure (G, L) and a subset K of G, we define the set of L-*representatives* of K to be the subset $L \cap \mu^{-1}(K)$ of $(S^{\pm})^*$ and the set of *reduced* L*-representatives* of K to be the subset $\bar{L} \cap \pi^{-1}(K)$ of $F(S)$; we observe that the images the two sets in G must both contain K, since L maps onto G.

We say that the subset K is *L-recognisable* if its L-representatives form a regular subset of $(S^{\pm})^*$ (equivalently, its reduced L-representatives form a regular subset of $F(S)$); we abbreviate $(S^{\pm})^*$-recognisable as recognisable.

Fig. 1 illustrates the set of L-representatives of a rational subset of a group. SR: I think this figure and caption need revision with π replaced by μ.

Following [6], we call a group G *automatic* if a rational structure (G, L) exists, which is additionally equipped with a collection of finite state automata $\{\mathcal{M}_s\}$ for $s \in S^{\pm} \cup \{1_G\}$ called *multipliers*, that accept strings corresponding to pairs of words (w_1, w_2) in $L \times L$ such that $w_1 s =_G w_2$ in the group.

Conventions for Cayley graphs and paths: Given a group $G = \langle S \rangle$, the *Cayley graph* of G with respect to S will be denoted by $\Gamma(G, S)$ (or just $\Gamma(G)$ if the generating set S is understood). The graph is directed and labelled. Its vertex set $V\Gamma$ is in correspondence with G and the label $l(v)$ of a vertex v is the corresponding element of G. For each $v \in V\Gamma$, and each $s \in S^{\pm}$, there is a directed edge $e \in E\Gamma \subseteq V\Gamma \times V\Gamma$ with label s, joining v to the vertex with label $l(v)s$.

By a path in the Cayley graph, we mean a directed graph homomorphism $\gamma : P_n \to \Gamma(G, S)$ from the path graph P_n into the Cayley graph. We often think of γ as its image in $\Gamma(G, S)$, rather than as a homomorphism.

For each edge $e = (u, v)$ in the Cayley graph labelled by a generator s, the graph has an inverse edge $e^{-1} = (v, u)$ with label s^{-1}; similarly, given a path γ in the graph from a vertex u to a vertex v we can assign to it a label $l(\gamma)$ in $(S^{\pm})^*$ given by the concatenation of the labels of edges in the path. Each such γ has a natural inverse path γ^{-1} whose label $l(\gamma^{-1})$ is the formal inverse $l(\gamma)^{-1}$ of $l(\gamma)$.

Since $V\Gamma$ is in bijection with G through the labelling function, we will often abuse notation and simply refer to a vertex v by its label $g = l(v)$. If the initial vertex for a path is fixed (e.g. the path starts at 1_G) the set of words in $(S^{\pm})^*$ is in bijection with the set of (not necessarily reduced) paths in the Cayley graph, so we will often interchangeably talk about a path γ or its label $w = l(\gamma)$.

For a path $\gamma : P_n \to \Gamma(G, S)$ with $l(\gamma) = w$ we denote by $|w|$ the length n of the path, which coincides with the word length of w in the free monoid; this notation coincides with the notation already introduced for monoids earlier in this section. Given an index $0 \le i \le |w|$, we denote by w_i the i-th vertex of the path and by $w(i)$ the prefix of w containing its first i letters, that is, $w(i) = \gamma|_{P_i}$. When γ is a path starting in 1_G, this means that $w_i = \mu(w(i))$.

Lastly, taking the usual graph metric where each edge has length 1 we can think of the Cayley graph of a group G, and therefore of G itself, as a metric space whose metric will be denoted by $d : G \times G \to \mathbb{N}$ throughout this text.

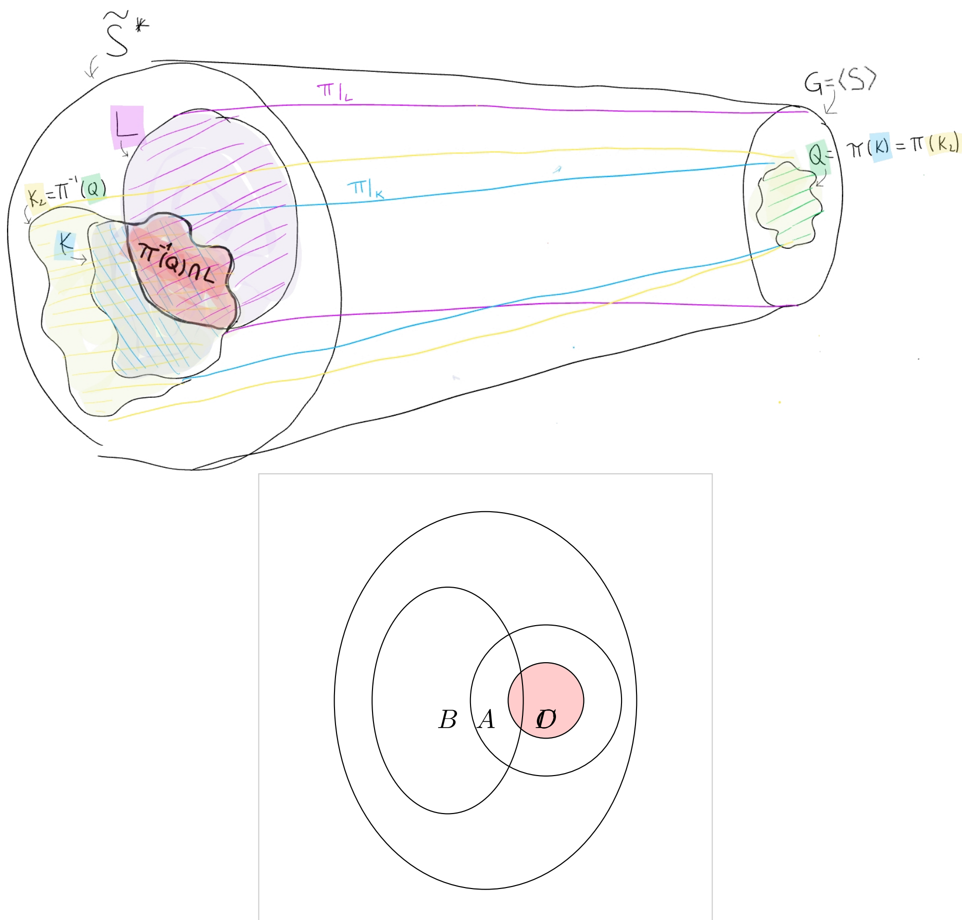

FIG. 1. L-representatives of a rational subset: L, Q are regular subsets of $(S^{\pm})^*$ and the projection $\pi\colon L \to G$ makes (G, L) a rational structure. The subset $K = \pi(Q)$ is rational in G, and its L-representatives are given by $P \cap L = \pi^{-1}(K) \cap L$. Note that, in general, $Q \cap L \neq P \cap L$. **LAM:** TODO make this better: less confusing, also $K \cap L \neq K_2 \cap L$. CHANGE KS AND QS, CHANGE Q2 FOR P.

Convexity notions: Taking into account the bijection between a group $G = \langle S\rangle$ and the vertices of the graph $\Gamma(G, S)$, it is natural to define a subset $K \subset G$ to be *convex* when, for any pair of elements $g, h \in K$, the vertices of any geodesic in $\Gamma(G, S)$ between g and h are all contained in K.

In practice, it is useful to work with a weaker condition than complexity: we define a subset K of G to be *quasi-convex* with constant k if each geodesic in $\Gamma(G, S)$ connecting g and h as above is contained in a k-neighbourhood of K [4, III.Γ.3]; we say that the subset is quasi-convex if such a constant k exists. In particular, the study of quasi-convex subgroups has been fruitful in the last decades (see [4, 5, 3]).

In the case where the group G has a rational structure $L \subset (S^{\pm})^*$, a more general concept is defined to work with subgroups of G: a subset K is said to be *L-quasi-convex* with constant k if, for any $g \in K$, every path in $\Gamma(G, S)$ from 1_G to g defined by a word in L (in other words, every L-representative of K) lies in the k-neighbourhood of K [7, 11]. Note that this definition is particularly powerful when K is a subgroup, as closure under inversion allows us to translate any path between elements g and h to a path between 1_G and the element $g^{-1}h$ in the subgroup. This is not the case for other subsets, e.g. rational subsets. In the case of subgroups, L-quasi-convexity is equivalent to L-recognisability [7].

Lastly, to point out some relations between quasi-convexity and L-quasi-convexity, let $L_{\texttt{geod}}$ be the language of geodesics in $\Gamma(G, S)$. If $L = L_{\texttt{geod}}$, then a subgroup is quasi-convex if and only if it is L-quasi-convex; if $L_{\texttt{geod}} \subseteq L$, then an L-quasi-convex subgroup must be quasi-convex; if $L \subseteq L_{\texttt{geod}}$ and K is a quasi-convex subset containing 1_G, then K must be L-quasi-convex.

As we noted, the definition of L-quasi-convexity is very useful when working with subgroups but a generalisation is needed to work with the rational subsets considered in this article: this motivates our definition of L-proximacy in Section 3.

3. Setup

Let (G, L) be a rational structure for a finitely presented group $G = \langle S | R\rangle$ and let $\mu\colon (S^{\pm})^* \to G$ and $\pi\colon F(S) \to G$ be the projections to G from the free monoid and from the free group, respectively. Given $Q \subseteq F(S)$ with $K = \pi(Q)$ a rational subset of G, our goal is to build an automaton that accepts the set of L-representatives of K .

$$
\begin{tikzcd}
Q \arrow[d, hook] \arrow[dr, "\pi"] & \\
F(S) \arrow[r, two heads, "\pi"] & G = \langle S|R\rangle \\
L \arrow[u, hook] \arrow[ur, hook, two heads, "\pi"] &
\end{tikzcd}
$$

AD: In the diagram above, why is L assumed to be subset of $F(S)$?

This will not always be possible (note that subgroups are rational subsets, and the subgroup membership problem is not always decidable), but if K is L-proximate (as defined in Definition 3.1) then there exists a procedure

that eventually produces such an automaton, as we will see in Section 4. If the rational subset is a submonoid given by a finite generating set, then a weaker hypothesis of weak proximacy will be sufficient (as defined in Definition 3.2)

We need to define L-proximacy for subsets K of G, but we need to start by defining it for the subsets Q of $(S^{\pm})^*$ that map onto them.

SR: rewrote defn, left in the terms 'fellow travel' etc., not sure if we wanted to do that

SR: I don't know what the reference to Fig 2 means, right at the beginning of this definition; that needs to be explained

Definition 3.1. Let (G, L) be a rational structure for a group $G = \langle S\rangle$, and let $Q \subseteq (S^{\pm})^*$. We say that Q is *L-proximate* in G with constants k, c if for every word w in $(S^{\pm})^*$ that is an L-representative of an element in $\mu(Q)$, there exists a word $w' \in Q$ with $\mu(w) = \mu(w')$ such that the paths in the Cayley graph of G based at 1_G and labelled w and w', *k-fellow travel up to at most c-reparametrisation*. That is:

there exists a function $f\colon [0,|w|] \to [0,|w'|]$ satisfying $f(0) = 0$, $f(|w|) = |w'|$ and $0 \le f(i+1) - f(i) \le c$, and such that, using the distance d in the Cayley graph $\Gamma(G,S)$, $d(w_i, w'_{f(i)}) \le k$ for all $0 \le i \le |w'|$. (See Figure Fig. 2.)

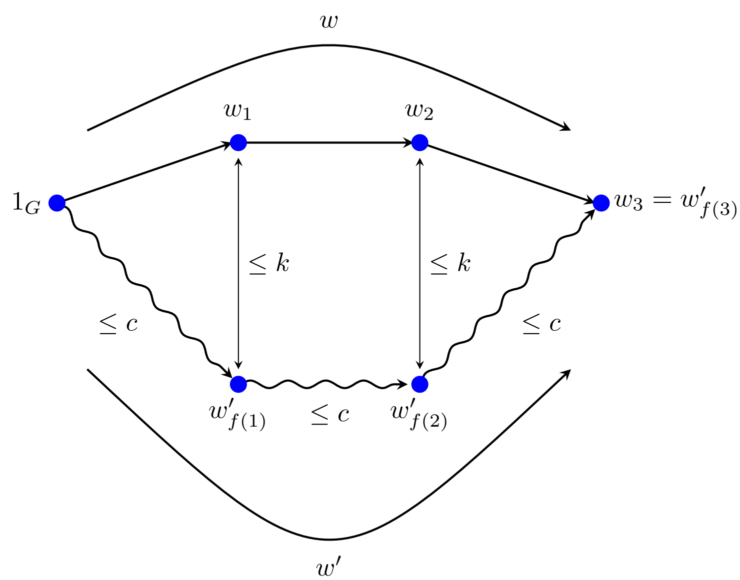

FIG. 2. Illustration of the paths in the Cayley graph defined in Definition 3.1.

In particular, w and w' *asynchronously fellow travel* (see [6] for a definition), with a (linear) restriction on their asynchronicity. A weaker notion of convexity, useful when working with submonoids of groups, is obtained when dropping the constant c:

Definition 3.2. Let (G,L) be a rational structure for a group $G = \langle S \rangle$, and let $Q \subseteq (S^{\pm})^*$. We say that Q is *weakly-L-proximate* in G with constant k if for every word w in $L \cap \mu^{-1}(\mu(Q))$ there is a word $w' \in Q$ and a map $f\colon [0,|w|] \to [0,|w'|]$ satisfying $\mu(w) = \mu(w')$, $f(0) = 0$, $f(|\bar{w}|) = |w'|$ and, using the distance d in the Cayley graph $\Gamma(G,S)$, $d(w_i, w'_{f(i)}) \le k$ for all $0 \le i \le |w'|$. In other words, the paths in the Cayley graph of G represented by w' and w must k-fellow travel asynchronously.

Remark 3.3. Note that we may have distinct subsets of $S^{\pm}$ that map onto the same subset of G, one of which is L-proximate (or weakly L-proximate) while the other is not. Take the group $\mathbb{Z}^2 = \langle a, b \mid [a,b] \rangle$ with rational structure $L = \{a, a^{-1}, b, b^{-1}\}^*$. The sets $Q_1 := L$ and $Q_2 := \{a^i b^j : i,j \in \mathbb{Z}\}$ both map onto G, but Q_1 is L-proximate while Q_2 is not. However, as we shall see in Corollary 3.11, two finitely generated submonoids of $(S^{\pm})^*$ that map to the same submonoid of G are either both (weakly) L-proximate or both not (weakly) L-proximate.

Recognising dependence on the choice of Q, we define L-proximacyfor a subset K of G as follows:

Definition 3.4. Given a rational structure (G, L) for a group $G = \langle S \rangle$ and a language $Q \subset (S^{\pm})^*$ which is *(weakly-)L-proximate* in G with constants k, c, we say that $K = \mu(Q)$ is *(weakly-)L-proximate* with respect to Q, with constants k, c.

AD: maybe this should be left as a remark about notation, rather than a definition

We dedicate the rest of this section to understanding the relationship between weakly-L-proximate, L-proximate and L-quasi-convex subsets. It is clear that L-proximate subsets are always weakly-L-proximate, and we shall prove below that a regular language Q that is weakly-L-proximate is also L-quasi-convex. Moreover, for submonoids given by a finite generating set T, if T^* is weakly-L-proximate it is also L-proximate and, for finitely generated subgroups, the three notions of convexity are equivalent. Briefly, we observe the following equivalences.

$$\begin{array}{rlclcl}
\text{f.g. subgroup } \pi(T^*) \subseteq G\colon & \text{L-qc} & \iff & \text{weakly-L-proximate} & \iff & \text{L-proximate} \\
\text{f.g. submonoid } \pi(T^*) \subseteq G\colon & \text{L-qc} & \Longleftarrow & \text{weakly-L-proximate} & \iff & \text{L-proximate} \\
\text{rational subset } \pi(Q) \subseteq G\colon & \text{L-qc} & \Longleftarrow & \text{weakly-L-proximate} & \Longleftarrow & \text{L-proximate}
\end{array}$$

Proposition 3.5, Proposition 3.6 and Proposition 3.7, which follow, provide detail and justification for these observations. Proposition 3.6 and Proposition 3.7 are illustrated by Fig. 3 and Fig. 4 respectively, but no further proof of either is provided.

SR: reworded text around the three propositions. In the end I put the proof of the first after its statement, realising that the figures already took up sopace that broke up the three statements, and the figures were anyway not necessarily placed next to the propositions.

Proposition 3.5. *Let $G = \langle S \rangle$ be a finitely generated group with rational structure (G, L) and let $Q \subset (S^{\pm})^*$ be a regular set. If Q is weakly-L-proximate in G with constant k then it is also L-quasi-convex with constant $k' = k + |V|$, where $|V|$ denotes the number of vertices in the minimal automaton accepting Q.*

Proof. Let w be a word in $L \cap \mu^{-1}(K)$ and let w' be the associated word in Q with which we are provided because of K being weakly-L-proximate. We know that, for $0 < i < |w|$, each vertex w_i is at distance at most k from the vertex $w'_{f(i)}$ in $\Gamma(G, S)$, and if the prefix words $w'(f(i))$ were in Q this would already show that K is L-quasi-convex. Even though this is not necessarily true, something very close to this does holds, as follows.. there exist words $u(i)$ for $0 < i < |w|$, each of length less than $|V|$, so that the concatenation $x(i) = w'(f(i))u(i)$ belongs to Q and therefore, if we see $x(i)$ as a path in $\Gamma(G, S)$ starting at 1_G, w_i is at distance at most $k + |V|$ from the final vertex of $x(i)$.

SR: reworded. I hope this is what Lucia intended. And a little minor rewording below.

To see that such a word $u(i)$ exists, let $\mathcal{A}_Q$ be the minimal, trim automaton associated to Q; note that each prefix word $w'(f(i))$ labels a path in $\mathcal{A}_Q$ starting at the initial state and finishing at a certain state v that is not necessarily accepting. Since $\mathcal{A}_Q$ is trim, there is a sequence of transitions in the automaton, starting at v and ending in an accepting state, and since $\mathcal{A}_Q$ has a finite number $|V|$ of states, this sequence of transitions can be chosen to be of length at most $|V|$. We just need to take $u(i)$ to be the word labelled by this sequence of transitions. □

Proposition 3.6. *Let $G = \langle S \rangle$ be a finitely generated group and (G, L) a rational structure as above. Let $M = \langle T \rangle$ be a submonoid with a finite generating set $T = \{t_1, \ldots, t_n\} \subset (S^{\pm})^*$ that is weakly-L-proximate in G, with constant k. Define* *constants l, c_1, c_2 as*

$$\begin{aligned} l &= \max\{|t|_S : \ t \in T\}, \\ c_1 &= \max\{|m|_T : \ m \in M \ \textit{and} \ |m|_S \leq 2(l+k)+1\}, \\ c_2 &= \max\{|m|_S : \ m \in T\}. \end{aligned}$$

AD: changed M to l in the first line of the definitions above, and removed the n's from the definitions of l, c_1, c_2.

The submonoid M, considered as the image under μ of the regular language T^*, is L-proximate with constants k, $c = c_1c_2$. (See Fig. 3.)

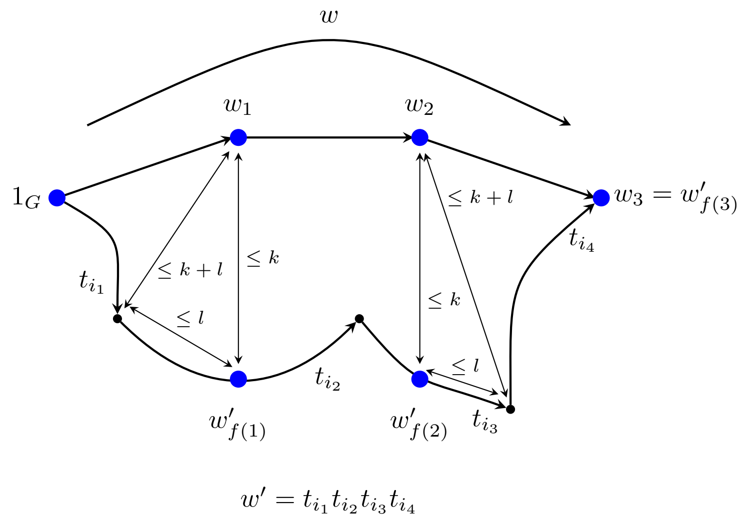

FIG. 3. Weakly-L-proximate submonoids are L-proximate. The distance in w' between $w'_{f(1)}$ and $w'_{f(2)}$ is bounded by the distance in w' between the initial vertex of $t_{i_2} \in T$ and the final vertex of $t_{i_3} \in T$. These vertices sit at most l edges along w' away from $w'_{f(1)}$ and $w'_{f(2)}$ respectively, so their distance in the S-metric is bounded by $2(k+l)+1$.

Proposition 3.7. *Let $G = \langle S \rangle$ be a finitely generated group and (G, L) a rational structure as above. Let $H = \langle T \rangle$ be a subgroup with a finite generating set $T \subset (S^{\pm})^*$. Define constants c_1, c_2 as*

$$\begin{aligned} c_1 &= \max\{|h|_T : \ h \in H \ \textit{and} \ |h|_S \leq 2k+1\}, \\ c_2 &= \max\{|h|_S : \ h \in T\}. \end{aligned}$$

If H is L-quasi-convex with constant k, then, considered as the image under μ of the regular language $Q = T^*$, H is L-proximate with constants k and $c = c_1c_2$. (See Fig. 4.)

At the end of Section 2 we pointed out that a subgroup is L-quasi-convex if and only if it is L-recognisable. Similarly, we have the following.

SR: replace 'we now have:' by 'we have the following.'

Proposition 3.8. *Given a rational structure (G, L) for a group, a rational subset K of G is L-proximate if and only if it is L-recognisable.*

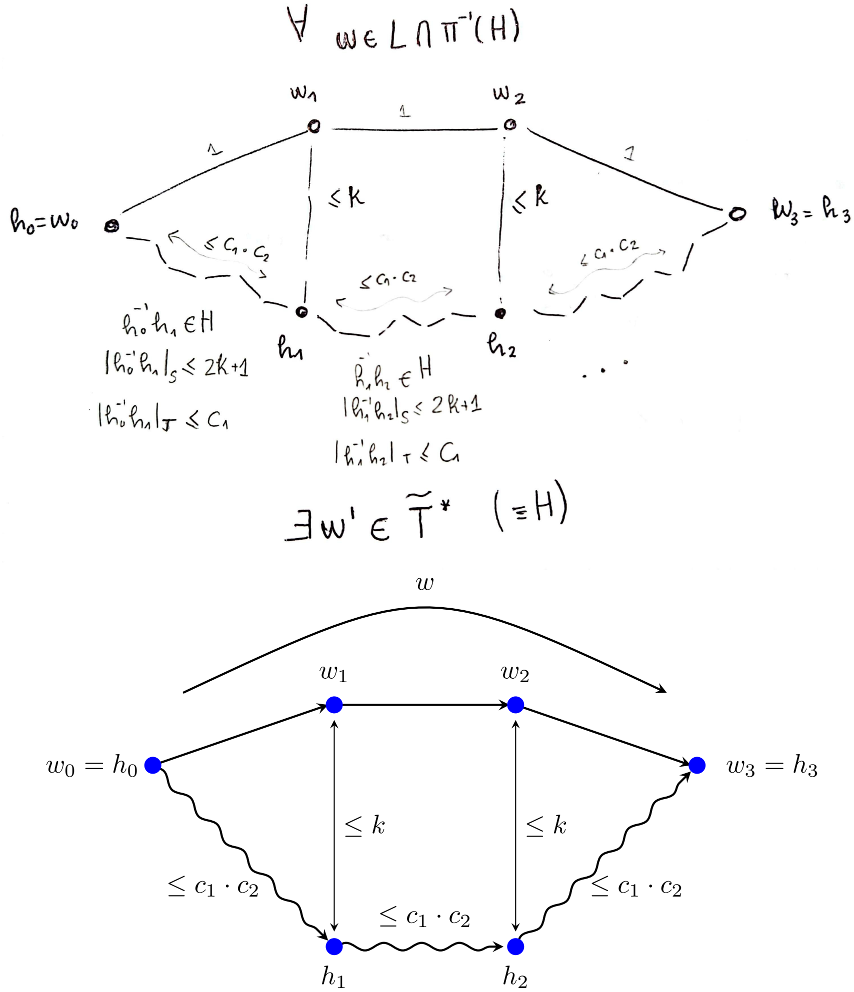

FIG. 4. L-quasi-convex subgroups are also L-proximate subsets of the group. Here $w \in L \cap \pi^{-1}(H)$, so by L-proximateness is in the k-neighbourhood of H, as shown by elements h_i. Then, for $i \in [0, 2]$, $h_i^{-1}h_{i+1} \in H$ and $|h_i^{-1}h_{i+1}|_S \leq 2k+1$ so $|h_i^{-1}h_{i+1}|_T \leq c_1$, whence the path shown between h_i and h_{i+1}, which is labelled by elements of $T^{\pm 1}$, has S-length at most c_1c_2. **AD:** at least I hope this is what's shown.

One direction is immediate, as L-recognisability means that $L \cap \mu^{-1}(K)$ is a regular language Q, and Q is by definition L-proximate with constants $k = 0$, $c = 1$. The other direction of the statement will be proved constructively in Proposition 4.9. As a consequence of this result and the above relationships, we have the following.

Proposition 3.9. *Given a rational structure (G, L) for a group, and a finitely generated submonoid T^* of G, if T^* is weakly-L-proximate then it is L-recognisable.*

We do not know if the converse holds. Before we move on to the next section, we give an equivalent definition for weakly-L-proximate submonoids that makes more clear the similarity to classical L-quasi-convexity and the independence on generating set.

Proposition 3.10. *Given a finitely generated group $G = \langle S \rangle$ with rational structure (G, L) and a finite generating set $T \subset (S^{\pm})^*$, the language T^* (and therefore the submonoid $M = \mu(T^*)$) is weakly-L-proximate if and only if there is a constant k such that for every $w \in L \cap \mu^{-1}(M)$ there are elements $m_1, \dots m_{|w|}$ in M such that, in $\Gamma(G, S)$, m_i is at distance at most k from the vertex w_i, $m_{|w|} = w_{|w|}$ and $m_{i+1} \in m_i M$.*

The equivalence is clear if we allow the constant k of quasi-convexity to change to $k' = k + l$, where l is defined as before by $l := \max\{n \colon n = |t|_S,\ t \in T\}$, and noting that the condition $m_{i+1} \in m_i M$, $m_{|w|} = w_{|w|}$ precisely allows us to "connect" the vertices m_i in a word $w' =_G w$.

The following is an immediate consequence of Proposition 3.10.

Corollary 3.11. *In a group with rational structure (G, L), if the images in G of finite subsets T and U of $(S^{\pm})^*$ generate the same submonoid of G, then T^* is (weakly) L-proximate precisely when U^* is.*

4. A folding algorithm for rational subsets

Throughout this section, we adopt the convention that a finitely presented group is given by a finite presentation $G = \langle S \,|\, R \rangle$ for which the set R of relators is closed under taking inverses and cyclic permutation of words.

4.1. **Rewriting.** In the next paragraphs we define a finite rewriting system, parallel and similar to the one presented in [11, §4], whose aim is to reflect the behaviour of the algorithm that will be introduced later in the section in Construction 4.6.

Let $G = \langle S \mid R \rangle$ be a finitely presented group and $\widetilde{D}_R$, $\widetilde{D}_{red}$ be the following rewriting systems on $(S^{\pm})^*$, where $R_0 = \{ss^{-1} \,|\, s \in S^{\pm}\}$:

$$\widetilde{D}_R = \{\varepsilon \to t \,|\, t \in R \cup R_0\},$$
$$\widetilde{D}_{red} = \{r \to \varepsilon \,|\, r \in R_0\}.$$

Given words w, w' in $(S^{\pm})^*$, we say that w' is one $\widetilde{D}_R$-rewriting step away from w if there exists an integer $n \geq 1$ and words $u^{(i)}$ such that w can be written as the concatenation

$$w = u^{(1)} u^{(2)} \dots u^{(n)},$$

(where some of the $u^{(i)}$ may be the empty word) and

$$w' = u^{(1)} t^{(1)} u^{(2)} \dots t^{(n-1)} u^{(n)}$$

so that, for each i, $\varepsilon \to t^{(i)}$ is a rule in $\widetilde{D}_R$. We denote this by $w \underset{R}{\to} w'$. Note that, by allowing $n = 1$ in the definition of a $\widetilde{D}_R$-rewriting step, we have that $w \underset{R}{\to} w$, and by allowing $u^{(i)} = \varepsilon$ we allow insertion of multiple relations in the same position in w and at its beginning or end. If there are words $w_{(1)}, \dots, w_{(k-1)}$ such that

$$w \underset{R}{\to} w_{(1)} \underset{R}{\to} \dots \underset{R}{\to} w_{(k-1)} \underset{R}{\to} w',$$

then we say that w is k $\widetilde{D}_R$-rewriting steps away from w' and we write $w \xrightarrow[R]{k} w'$. We write $w \xrightarrow[R]{*} w'$ to describe that w' is obtained from w in 0 or more $\widetilde{D}_R$-rewriting steps. Using the rewriting system $\widetilde{D}_{red}$, we similarly define $\widetilde{D}_{red}$-rewriting steps together with the notation

$$w \xrightarrow[red]{} w' \ , w \xrightarrow[red]{k} w' \ \text{ and } \ w \xrightarrow[red]{*} w'.$$

Lemma 4.1. *If w, u and w' are elements of $(S^{\pm})^*$ such that $w \xrightarrow[red]{} u \xrightarrow[R]{} w'$, then there exists a word $v \in (S^{\pm})^*$ such that $w \xrightarrow[R]{} v \xrightarrow[red]{} w'$.*

Rather than proving this in general we exhibit an example. The (rather tedious) general proof follows the same lines. For example, suppose $w = ass^{-1}btt^{-1}c$, where $s, t \in S$ and $a, b, c \in (S^{\pm})^*$, with $u = abc$ and then that $u = defg$, where $d, e, f, g \in (S^{\pm})^*$, and $w' = d\alpha e\beta f\gamma g$, where $\alpha, \beta, \gamma \in R \cup \{ss^{-1} \mid s \in S\}$. Then we may refine the factors of u to allow either factorisation. If, for instance, $a = a_1a_2$ and $b = b_1b_2$ where $d = a_1$, $e = a_2b_1$, $f = b_2$ and $g = c$ then

$$u = (a_1a_2)(b_1b_2)c = a_1(a_2b_1)b_2c,$$

$$w = (a_1a_2)ss^{-1}(b_1b_2)tt^{-1}c = a_1(a_2ss^{-1}b_1)(b_2tt^{-1})c$$

and

$$w' = a_1\alpha(a_2b_1)\beta b_2\gamma c.$$

Defining $v = a_1\alpha a_2ss^{-1}b_1\beta b_2tt^{-1}\gamma c$ we then have $w \xrightarrow[R]{} v \xrightarrow[red]{} w''$, as required.

As a consequence of Lemma 4.1:

Lemma 4.2. *If w, w' are words representing the same element in G, then w' can be obtained from w by applying a sequence of $\widetilde{D}_R$ rewrites followed by $\widetilde{D}_{red}$ rewrites.*

Definition 4.3. Given a finitely presented group $G = \langle S|R\rangle$, and words w, w' in $(S^{\pm})^*$ we say that w' is one step away from w if there exists an intermediate word w'' such that

$$w \xrightarrow[R]{} w'' \xrightarrow[red]{*} w'.$$

We say that w' is k steps away from w if there are words $w_{(0)} = w, w_{(1)}, \dots, w_{(k)} = w'$ such that $w_{(i)}$ is one step away from $w_{(i-1)}$ for $i = 1, \dots, k$.

Lemma 4.4. *Let w and w' be words in $(S^{\pm})^*$. Then w' is k steps away from w if and only if there exists an intermediate word w'' such that*

$$w \xrightarrow[R]{k} w'' \xrightarrow[red]{*} w'.$$

Proof. If w' is k steps away from w then, by repeatedly applying the result of Lemma 4.1 a rewriting of the required form may be obtained. The converse follows directly from the definition. $\square$

4.2. **The procedure.** Now, we describe the procedure that will allow us to recognise all of the L-representatives of a rational subset K of a group $G = \langle S\rangle$, starting with an automaton that recognises a rational subset Q of $(S^{\pm})^*$ with $\mu(Q) = K$. SR: inserted formal definition of folding

A key step in our procedure *folds* a finite state automaton $\mathcal{A} = (S^{\pm}, V, E, A, v_0)$ to produce a finite state automaton $\mathcal{A}'$ whose language is the *folding* of $L(A)$.

Definition 4.5. The *folding* of a subset L of $(S^{\pm})^*$ is the smallest subset L' of $(S^{\pm})^*$ containing L for which, if $uxx^{-1}v \in L'$ then $uv \in L'$.

By Benois' theorem [2, 1], the folding L' of a regular language $L = L(\mathcal{A})$ is regular, and an automaton $\mathcal{A}'$ to recognise it can be constructed using an algorithm described in [1]. We call the automaton $\mathcal{A}'$ the *folding* of $\mathcal{A}$.

AD: If $\mathcal{A}$ is trim so is $\mathcal{A}'$.

AD: Give a bried description of Bartholdi and Silva's algorithm.

Construction 4.6. Let $G = \langle S \mid R \rangle$ be a finitely presented group, with R closed under taking inverses, and let $\mathcal{A}_0$ be an automaton over the alphabet $S^{\pm}$ recognising a regular language $Q \subseteq (S^{\pm})^*$. For each $n > 0$, we construct $\mathcal{A}_n$ from $\mathcal{A}_{n-1}$ by performing the following sequence of two steps:

(1) At each vertex of $\mathcal{A}_{n-1}$ add a cycle corresponding to each relator of R as well as a cycle of length two corresponding to each ss^{-1} for $s \in S^{\pm}$, and hence obtain an automaton $\mathcal{A}_n^0$.
(2) Use the algorithm of [1] to construct an automaton $\mathcal{A}_n^1$ by folding $\mathcal{A}_n^0$.
(3) Determinise $\mathcal{A}_n^1$ to obtain A_n.

iLAM: The following prop is a formulation in terms of rewriting systems, as andrew wanted. unsure if its better than the original (january25) version of it. I think Ive rewritten proof correctly, hopefully no big mess made.

Proposition 4.7. *A word $w \in (S^{\pm})^*$ is recognised by the automaton $\mathcal{A}_n$ if and only if w is at most n steps away from a word accepted by $\mathcal{A}_0$.*

Proof. By definition, $\mathcal{A}_n$ is the automaton obtained by folding and determinising $\mathcal{A}_n^0$, which means that the language accepted by $\mathcal{A}_n$ is the set of all words that can be obtained through free reductions from words accepted by $\mathcal{A}_n^0$. Consequently, a word w' is accepted by $\mathcal{A}_n$ if and only if there is a word w'' accepted by $\mathcal{A}_n^0$ such that $w'' \xrightarrow[red]{*} w'$.

Take now an arbitrary word w'' accepted by $\mathcal{A}_n^0$. It is clear from the construction that we can write it as

$$w'' = r^{(0)} w_1 r^{(1)} w_2 \dots w_m r^{(m)},$$

where each word $r^{(i)}$ is in the language $(R \cup R_0)^*$ (where $R_0 = \{ss^{-1} \mid s \in S^{\pm}\}$, as above) and $w = w_1 w_2 \dots w_m$ is a word of length m accepted by $\mathcal{A}_{n-1}$. This means exactly that w'' is at most one R-rewriting step away from w, so altogether we deduce that

$$w \xrightarrow[R]{} w'' \xrightarrow[red]{*} w'.$$

This means that w' in $L(\mathcal{A}_n)$ is one step away from a word w in $L(\mathcal{A}_{n-1})$ and, by induction, we have that the words accepted by $\mathcal{A}_n$ are at most n rewriting steps away from those accepted by $\mathcal{A}_0$.

The converse also follows by induction from the definition of the rewriting systems, since $\widetilde{D}_R$ emulates step (1) and $\widetilde{D}_{red}$ emulates step (2) in Construction 4.6: if a word w' is at most one step away from a word w accepted by $\mathcal{A}_{n-1}$ it means that we can find a word w'' (possibly equal to w) such that

$$w \xrightarrow[R]{} w'' \xrightarrow[red]{*} w'.$$

Applying Construction 4.6, we see that the word w'' is accepted by $\mathcal{A}'_{n-1}$ and the word w' is accepted by $\mathcal{A}_n$. $\square$

Corollary 4.8. *Let $\langle S|R \rangle$ be a finitely presented group and $Q_0 \subseteq (S^{\pm})^*$ the regular language accepted by some automaton $\mathcal{A}_0$. Denote by Q_n, $n \in \mathbb{N}$, the language accepted by the automaton $\mathcal{A}_n$ whose construction is described above. Then, $\bigcup_{n \in \mathbb{N}} Q_n = \pi^{-1}(\pi(Q_0))$.*

Proof. From Proposition 4.7, we see that $\pi(Q_n) = \pi(Q_0)$ for all $n \in \mathbb{N}$, and it is clear that for any $w \in \pi^{-1}(\pi(Q_0))$, there exists $n \in \mathbb{N}$ such that w can be obtained in n steps from a word of Q_0. □

When dealing with L-proximate rational subsets, there exists some $n \in \mathbb{N}$ such that Q_n contains all of $\pi^{-1}(\pi(Q_0)) \cap L$. Indeed:

Proposition 4.9. *Let (G, L) be a rational structure for a finitely presented group $G = \langle S|R\rangle$ and let $Q_0 \subseteq (S^{\pm})^*$ be the regular language accepted by an automaton $\mathcal{A}_0$. Assume that Q_0 is L-proximate with constants c, k (see Definition 3.1) and let $K = \pi(Q_0)$. Then, there exists $n \in \mathbb{N}$ such that the full set of L-representatives of K is contained in Q_n, the language accepted by the automaton $\mathcal{A}_n$ whose construction is described above.*

The following lemma will aid the proof of Proposition 4.9.

Lemma 4.10. *Let $G = \langle S\,|\,R\rangle$ be a finitely presented group, let $m, n \in \mathbb{N}$ be strictly positive integers, and let $w_1, \ldots, w_m, w'_1, \ldots, w'_m \in (S^{\pm})^*$ be words such that each w'_i can be obtained in at most n steps from w_i. Then, $w'_1 w'_2 \ldots w'_m$ can be obtained in n steps from $w_1 w_2 \ldots w_m$.*

Proof. The result follows directly from the definitions: it is sufficient to prove the result for $m = 2$, and by induction, it suffices to show it for $n = 1$, which is immediate. □

Proof of Proposition 4.9. Let $n \in \mathbb{N}$ be defined as follows:

$$n = \max\{m \colon u \text{ is } m \text{ steps away from } v \text{ for } \; u, v \in (S^{\pm})^* \text{ with } \\ u =_G v,\ |u| \leq 2k+1,\ |v| \leq c\}$$

The numbers m exist by Lemma 4.1 and, since n is obtained by studying a bounded ball in the Cayley graph of G with respect to S, it is clear that the maximum is reached and n is well defined. We will show that the set of L-representatives $J = L \cap \pi^{-1}(K)$ is a subset of Q_n.

Let w be an arbitrary word in the language J. By assumption, there exists a word $w' \in Q_0$ with $w =_G w'$ such that the paths in the Cayley graph of G corresponding to the words w and w' k-fellow travel, up to c-reparametrisation. Let $f\colon [0, |w|] \to [0, |w'|]$ be the reparametrisation map so that, for the paths labelled by w and w' and following the notation defined in Section 2 , we have $d(w_i, w'_{f(i)}) \leq k$ for all $0 \leq i \leq |w|$.

See Fig. 5 for an illustration of the following paragraph. Following Section 2 denote by $w(1), \ldots, w(|w|)$ the letters in w, which we can think of as S-labelled, 1-edge subpaths of w in the Cayley graph. Let $\gamma(1), \gamma(2), \ldots, \gamma(|w|)$ be the subwords of w' defined by respectively taking its first $f(1)$ letters, then the next $f(2) - f(1)$ letters, etc. until finally making $\gamma(|w|)$ out of the last $f(|w|) - f(|w| - 1)$ letters of w'. Each $\gamma(i)$ can be thought of as a subpath of w' in the Cayley graph of length less than or equal to c. Now, let $\delta(i) \in F(S)$ be a word representing a geodesic path between the vertices w_i and $w_{f(i)}$, for $i = 0, \ldots, |w|$. Note that such words have length less or equal than k, and that $\delta(0), \delta(|w|)$ are trivial paths. Lastly, define $\alpha(i)$ to be the path in the Cayley graph (and the word in $(S^{\pm})^*$) defined by concatenating the paths $\delta^{-1}(i-1) \cdot w(i) \cdot \delta(i)$, where δ^{-1} denotes the natural inverse path. The paths $\alpha_1, \ldots, \alpha_{|w|}$ all have length less than or equal to $2k + 1$.

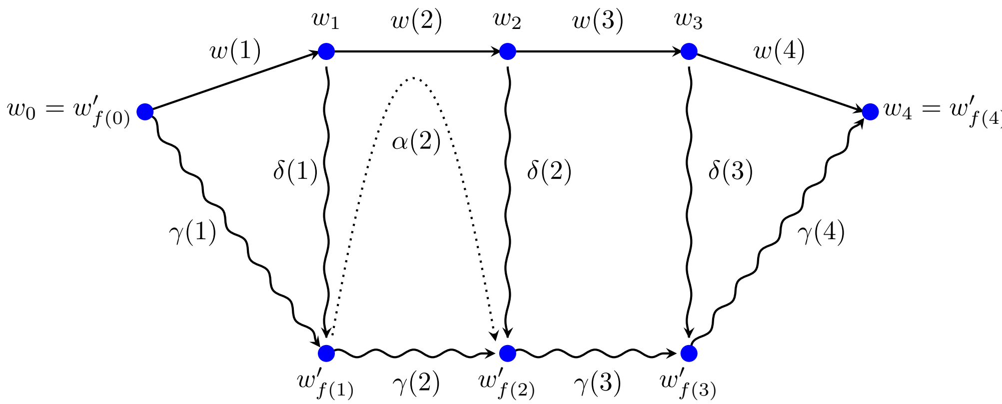

FIG. 5. Notation for the proof of Proposition 4.9

Note that each of the paths $\alpha(i)$ defines a word equivalent to $\gamma(i)$ in G, and because $|\alpha(i)| \leq 2k+1$ and $\gamma(i) \leq c$ the word $\alpha(i)$ is at most n steps away from $\gamma(i)$. By Lemma 4.10 we conclude that the word defined by $\alpha = \alpha(1) \cdot \ldots \cdot \alpha(|w|)$ is at most n steps away from $w' = \gamma(1) \cdot \ldots \cdot \gamma(|w|)$ and, since w' is in Q_0, this means that Q_n contains the word defined by α. Furthermore, $\alpha = \alpha(1) \cdot \ldots \cdot \alpha(|w|)$ is freely reducible to $w = w(1) \cdot \ldots \cdot w(|w|)$ which means that w is also accepted by Q_n (since its language is closed under free reductions). $\square$

Definition 4.11. Given a set T of words over $S^{\pm}$, we define the *flower automaton* $\mathsf{Flower}(T)$, which accepts T^*, to be the automaton over $S^{\pm}$ formed by attaching disjoint directed loops, each labelled by an element $t \in T$ to a vertex s_0 that is the both the start state and the single accepting state.

SR: added def of $\mathsf{Flower}(T)$. I think it is supposed to accept T^* and not $(T^{\pm})^*$. But I'd like to clarify this.

As a consequence of Proposition 4.9 and Proposition 3.6, we have the following result.

Proposition 4.12. *Let (G, L) be a rational structure for a finitely presented group $G = \langle S|R \rangle$, let $T \subseteq (S^{\pm})^*$ be a finite set of words and consider the regular language $Q_0 := T^*$ accepted by the flower automaton $\mathcal{A}_0 := \mathsf{Flower}(T^*)$ (Definition 4.11). If the submonoid $M = \pi(Q_0)$ is weakly-L-proximate with constant k, and with respect to Q_0 (see Definition 3.2), then there exists $n \in \mathbb{N}$ such that the full set of L-representatives of M is contained in Q_n, the language accepted by the automaton $\mathcal{A}_n$ whose construction is described above.*

5. Application to submonoids of automatic groups

AD: process procedure to produce algorithm?

In this section we use the argument of Kharlampovich, Myasnikov and Weil [11] to turn the process of the previous section into a procedure which will construct an automaton accepting the set of L-representatives of a given finitely generated monoid of an automatic group. The main ingredient is a test to show whether or not the process has completed. The test works for submonoids because the automaton corresponding to a finite generating set of a monoid has start state equal to its unique accept state. As we shall see, this means that the following lemma, from [11], gives rise to the required test. Throughout the section, the conventions for symmetric presentations of a group remain as in Section 4.

For the remainder of this section let G be an automatic group with generating set S and $L \subseteq (S^{\pm})^*$ be the language of this automatic structure.

Lemma 5.1 (Lemma 4.11 in [11])**.** *Given $h \in F(S)$ and a regular language Q contained in L, there exists an algorithm that constructs an automaton accepting the set $\mathcal{M}_h(Q) = L \cap \pi^{-1}(\pi(Qh))$ of all the L-representatives of the elements of $\pi(Qh)$.*

Proposition 5.2. *Let T be a finite subset of $S^{\pm}$, such that the submonoid $M = \pi(T^*)$ of G generated by T is weakly-L-proximate. Let $\mathcal{A}_0 := \mathsf{Flower}(T)$ (Definition 4.11), and let $\mathcal{A}_0, \mathcal{A}_1, \dots$ be the sequence of automata constructed in Section 4. Then, for $i \geq 0$, the automaton $\mathcal{A}_i$ may be effectively constructed and there is an algorithm which, on input T, outputs an integer $n \geq 0$ such that the language $L(\mathcal{A}_n)$ contains all L-representatives of M.*

Proof. First we describe the algorithm and then we verify that it works as required. The algorithm begins by finding an element $w_\varepsilon \in L$ such that $\pi(w_\varepsilon) = 1_G$. As G is automatic this can be done using the multiplier automaton for ε. After this pre-processing step, for each $i \geq 0$ until it halts, the algorithm consists of 3 steps. The input at each iteration is the automaton $\mathcal{A}_i$ and we set $Q_i := L(A_i)$. In the first step the algorithm checks to see that w_ε is in Q_i. If not then the algorithm passes to the third step described below. If $w_\varepsilon \in Q_i$ the algorithm carries out its second step in which it

- constructs an automaton for the language $Q_i' := L \cap Q_i$;
- constructs an automaton accepting $\mathcal{M}_t(Q_i')$, for each $t \in \{\varepsilon\} \cup T$;
- tests to see whether or not $\mathcal{M}_t(Q_i')$ is contained in Q_i.

If the answer to the test is "yes", for all $t \in \{\varepsilon\} \cup T$, the algorithm halts and outputs i and $\mathcal{A}_i$. Otherwise the algorithm continues to step 3 in which Construction 4.6 is used to construct $\mathcal{A}_{i+1}$; and then moves on to iteration $i + 1$.

From Proposition 4.12, Lemma 5.1 and the standard theory of finite state automata, it follows that this procedure may be executed, and effectively constructs $\mathcal{A}_i$ at the ith iteration. To complete the proof we show that it halts and that, when it does so, the final claim holds.

First suppose that, for some $n \geq 0$, the language $Q_n := L(\mathcal{A}_n)$ contains $L \cap \pi^{-1}(M)$. Then, in particular, $w_\varepsilon \in Q_n$ and $L \cap \pi^{-1}(M) \subset L \cap Q_n = Q_n'$. As $\pi(Q_n') = \pi(Q_n) = M$ and $\pi(t) \in M$, for all $t \in \{\varepsilon\} \cup T$, we have $\pi(Q_n' t) \subseteq M$ so $L \cap \pi^{-1}(\pi(Q_n' t)) \subseteq L \cap \pi^{-1}(M) \subseteq Q_n$. Hence $\mathcal{M}_t(Q_n') \subseteq Q_n$, for all $t \in \{\varepsilon\} \cup T$.

Conversely suppose that, for some $n \geq 0$, $w_\varepsilon \in Q_n$ and, for all $t \in \{\varepsilon\} \cup T$, we have $\mathcal{M}_t(Q_n') \subseteq Q_n$. Suppose $w \in Q_0$, so $w = t_1 \cdots t_m$, where $t_i \in T$, $m \geq 0$. If $m = 0$ then $w = \varepsilon$. As $w_\varepsilon \in Q_n'$, we have $Q_n' \neq \emptyset$, so all L-representatives of $\pi(\varepsilon) = \pi(w_\varepsilon)$ belong to $\mathcal{M}_\varepsilon(Q_n') \subseteq Q_n$.

Now assume that $m \geq 1$, and that if u is a product of fewer than m elements of T, all L-representatives of $\pi(u)$ belong to Q_n. Then the L-representatives of $\pi(t_1 \cdots t_{m-1})$ (of which there is at least one) belong to Q_n', so the L-representatives of $\pi(w)$ all belong to $\mathcal{M}_{t_n}(Q_n') \subseteq Q_n$. As $\pi(Q_0) = M$, it follows that $L \cap \pi^{-1}(M) \subseteq Q_n$ if and only if both $w_\varepsilon \in Q_n$ and $\mathcal{M}_t(Q_n') \subseteq Q_n$, for all $t \in \{\varepsilon\} \cup T$. Therefore the algorithm halts with the required output. ∎

Remark 5.3. To obtain an automaton accepting exactly $L \cap \pi^{-1}(M)$ we use a standard algorithm for an automaton for the language $L(\mathcal{A}_n) \cap L$, where $\mathcal{A}_n$ is output from the algorithm above.

AD: needed/not needed?

Remark 5.4. The proof above breaks down if Q_0 is an arbitrary regular subset of $(S^{\pm})^*$. The difficulty is to find an analogue of the generating set T and then an analogue of the statement that given that $w_\varepsilon \in Q_n$ then the fact that $\mathcal{M}_t(Q'_n) \subseteq Q_n$, for all t, implies that $L \cap \pi^{-1}(M)$ is contained in Q_n. Both of these difficulties arise when Q_0 has distinct start and final state. In the first author's thesis further analysis of the general case of L-proximate rational subsets is carried out and some (technical) conditions are identified, under which a halting test can be devised.

6. Application to submonoids of surface groups

The goal of this section is to give examples of weakly-L-proximate submonoids that lie inside surface groups, using tools from small cancellation theory to ensure that the necessary convexity conditions hold.

SR: renamed gen set S_g as Y_g so that S_g can be the surface of genus g

Definition 6.1. The *surface group* Σ_g of genus $g \geq 1$ is the group with presentation

$$\Sigma_g \equiv \langle a_1, b_1 \dots a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] = 1 \rangle.$$

We shall write Y_g for the set of generators in the presentation above, and r_g for the single relator. The group Σ_g is the fundamental group of a closed, compact, orientable surface S_g of genus g. Except where otherwise stated, we assume that $g > 1$ throughout this section. In this case Σ_g is a hyperbolic group and therefore automatic (with the rational structure given by geodesics with respect to any finite generating set).

Given a set R of relators over a generating set S, the *symmetrisation* R_* is the set of cyclic permutations of words in R, together with their inverses. We write R_g for the symmetrisation of $\{r_g\}$. Recall that Dehn's rewriting system for Σ_g has the following rewriting rules:

(1) $ss^{-1} \to \varepsilon$, for a letter s in $Y_g^{\pm}$.
(2) $u \to v$, for words u, v in $(Y_g^{\pm})^*$ such that $uv^{-1} \in R_g$ and $|u| > |v|$.

A theorem of Dehn states that any word in $Y_g^{\pm}$ which represents the identity of Σ_g can be reduced to ε by repeated applications of these rewriting rules (Dehn's algorithm).

It is useful to think of the Cayley graph $\Gamma(\Sigma_g, Y_g)$ as a tessellation of the hyperbolic plane $\mathbb{H}^2$ by $4g$-gons with edges labelled by the relator r. The second of Dehn's rewriting rules replaces any path going around more than half of a $4g$-gon with the shorter route using the complementary edges. See Fig. 6 for some genus-2 examples.

6.1. Results from small cancellation theory.

We shall need to introduce a result, Theorem 9.4, from [13]. We recall some standard definitions from small cancellation theory, which can be found in [12], or in the small cancellation theory appendix to [8].

Definition 6.2. Let $P = \langle S \mid R \rangle$ be a group presentation. A word u over S is a *piece* with respect to P if u is a common prefix of two distinct relators in R_*.

For the standard presentation of Σ_g, the pieces are precisely the 1-letter words.

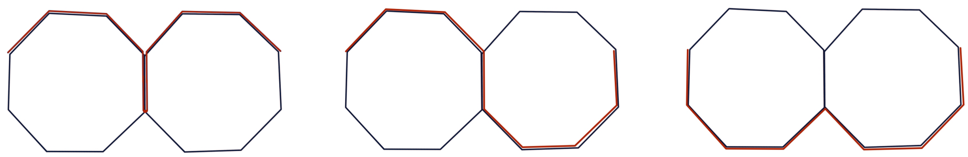

FIG. 6. Three examples of words in Σ_2 depicted as paths (in red) along the Cayley graph of Σ_2, which tessellates the hyperbolic plane with octagons. The leftmost path is not Dehn-reduced because it exhibits backtracking, the middle one is not Dehn-reduced because it goes around 5 edges of an octagon. The rightmost path represents a Dehn-reduced word.

Definition 6.3. Let $P = \langle S \mid R \rangle$ be a group presentation.

- For a natural number p, we say that a P satisfies condition $\mathrm{C}(p)$ if every relation in R_* is the concatenation of at least p pieces.
- For $q > 2$ we say that P satisfies condition $\mathrm{T}(q)$ if for every $\ell \in \{3, 4, \ldots, q-1\}$, and for every sequence $\{r_1, r_2, \ldots r_\ell\}$ in R_* such that $r_i \neq r_{i+1}^{-1}$ for all i, and such that $r_\ell \neq r_1^{-1}$, at least one of the products $r_1 r_2, \ldots, r_{\ell-1} r_\ell, r_1 r_{l-1}$ is freely reduced.

A presentation which satisfies both of the conditions $\mathrm{C}(p)$ and $\mathrm{T}(q)$ is said to be $\mathrm{C}(p)$–$\mathrm{T}(q)$. We note that the standard presentation for Σ_g is $\mathrm{C}(4g)$–$\mathrm{T}(4g)$; in particular, it follows that this presentation is $\mathrm{C}(4)$–$\mathrm{T}(4)$.

Recall that a Van Kampen diagram for the presentation $\langle S \mid R \rangle$ is a finite connected cell complex, embedded into $\mathbb{R}^2$, such that the edges are directed and labelled by elements of S, and such that the labels along the boundary of each 2-cell spell a element of R_*.

An *arc* in the Van Kampen diagram X is a simple path whose internal vertices have degree 2; an arc is *maximal* if it is not properly contained in any other arc. An arc is an *internal arc* if its interior lies in the interior of X, and a *boundary arc* if it lies entirely on the boundary of X. We notice that the labels along a internal maximal arc in a Van Kampen diagram spell a maximal piece. Hence for the usual presentation of Σ_g, all internal maximal arcs have length 1. (That is to say, there are no internal vertices with degree 2.)

The following definitions are taken from [13]; we note that the diagrams used there are topological disc diagrams (which include Van Kampen diagrams), so we have adopted the "informal" definitions given in the introduction, which apply more naturally to Van Kampen diagrams.

Definition 6.4. A *spur* is a 1-cell which is not in the boundary of a 2-cell and which is attached to the rest of the diagram at only one end. An *i-shell* is a 2-cell with exactly i maximal arcs of its boundary lying in the interior of the diagram.

For the next definition we follow [15].

Definition 6.5. A *ladder* L is a diagram which is the union of a sequence of closed 1-cells and 2-cells $c_1, \ldots, c_n$, such that for $1 < j < n$, there are exactly two components in $L - c_j$, and exactly one component in $L - c_1$ and $L - c_n$, and such that any c_i which is a 1-cell is not contained in any other closed c_j.

A ladder can be considered as a diagram with two distinguished points x and y on the boundary, such that if P and Q are the two boundary paths from x to y, then every 2-cell has at least one edge from P and another from Q, and such that every edge not contained in a 2-cell is common to both P and Q. (We consider the paths P and Q as the rails of the ladder.)

The following result is [13, Theorem 9.4] (stated there for disc diagrams, which include Van Kampen diagrams).

Theorem. If X is a Van Kampen diagram for a C(4)–T(4) presentation, then one of the following holds:

(1) X contains at least 3 spurs or i-shells with $i \leq 2$;
(2) X is a ladder;
(3) X consists of a single vertex or a single 2-cell.

6.2. **Dehn-reduced words and geodesics in surface groups.** Note that an arbitrary word w of $(Y_g^{\pm})^*$ is not necessarily reduced to a geodesic word by Dehn's algorithm. The rightmost path in Fig. 6 represents a Dehn-reduced word which is not geodesic. We shall show, using small cancellation properties of our presentation for Σ_g, that Dehn-reduced words asynchronously fellow-travel with geodesics. This gives us a tool to find weakly-L-proximate languages which describe submonoids of surface groups, and hence a class L-recognisable submonoids.

We need the following lemma.

Lemma 6.6. *Let X be a Van Kampen diagram for the surface group Σ_g. Suppose that the boundary cycle of X starting at a given vertex is labelled by a concatenation of Dehn-reduced words $w_1, \ldots, w_n$. Suppose also that X has an i-shell Y with $i \leq 2g-1$. Then the maximal arc which is the intersection of Y with the boundary of X, contains edges from two consecutive words $w_j w_{j+1}$ or $w_n w_1$.*

Proof. Since the only relator for Σ_g has length $4g$, it follows that the 2-cell Y has $4g$ edges, all of which are maximal arcs. Since at most $2g-1$ of these edges are internal, we see that Y has at least $2g+1$ boundary edges, and that the labels for these edges spell a subword of an element of R_g. But since each w_i is Dehn-reduced, none can contain a subword of an element of R_g of length $2g+1$, and so the boundary edges of Y must contain labels from two consecutive words $w_i w_{i+1}$. □

We can now obtain the following result.

Lemma 6.7. *Let u be a Dehn-reduced word in $(Y_g^{\pm})^*$, and let v be a geodesic representing the same element of Σ_g. (So v is also Dehn-reduced.) A Van Kampen diagram for the word uv^{-1} must be a single vertex, a single 2-cell, or a ladder. In particular, u asynchronously $(g-1)$-fellow travels with v.*

Proof. We see from [15, Corollary 2.8] that if X is a diagram for a C(4)–T(4) presentation, and if uv^{-1} is the boundary cycle of X, then either u or v contains all of the boundary edges of a 1-shell or a 2-shell of X, or else X is a single vertex, a 2-cell, or a ladder. Let X be a Van Kampen diagram for uv^{-1}, and let u_0 and v_0 be the vertices corresponding to the start of the words u and v^{-1} respectively. By Lemma 6.6 any 1- or 2-shell of X must include edges from both u and v^{-1}, and so neither u nor v^{-1} contains all of the boundary edges for for any 1- or 2-shell. So X is a vertex, a 2-cell, or a ladder.

If X is a vertex or a 2-cell, then clearly u and v fellow-travel, so we assume that X is a ladder. In this case its endpoints are u_0 and v_0, and the rails are u and v. Each cell c_i forming the ladder must intersect both rails, and this observation, together with the fact that shells along the Dehn-reduced word u intersect the boundary of X in at most $2g$ edges, shows that every vertex of u is at distance less than g from a vertex of v. □

6.3. **Constructing L-recognisable submonoids of surface groups.** We start with an immediate corollary to Lemma 6.7.

Lemma 6.8. *A Dehn-reduced language Q over the generators of Σ_g is weakly-L-proximate in the surface group Σ_g. In particular, if T is a generating set for a submonoid $M \subset \Sigma_g$, and if every word in T^* is Dehn-reduced, then M is L-recognisable, and therefore M has decidable membership problem.*

Before we give examples of submonoids to which this result applies, we note an immediate way to strengthen it.

Definition 6.9. Let G be a group with presentation $\langle S|R\rangle$, and let u, v be words in $(S^{\pm})^*$. We say that u yields v in a *simultaneous Dehn-reduction*, $u \xrightarrow[\text{sim}]{} v$, if we can write u and v as a concatenation of subwords $u = u_1 \dots u_n$ and $v = v_1 \dots v_n$ such that v_i can be obtained from u_i by applying a Dehn rewriting step. We say that u is N *simultaneous Dehn rewriting steps* from v.

Proposition 6.10. *Let Q be a language over Y_g all of whose words are at most N simultaneous Dehn rewriting steps away from being Dehn-reduced. Then Q is weakly-L-proximate with constant $k = (g-1)(N+1)$. If $Q = T^*$, then the submonoid $M = \pi(T^*)$ is L-recognisable and has decidable membership problem.*

Let u, v be words over Y_g. We note that if u is a single simultaneous Dehn rewriting step away from v, then every vertex from v in a Van Kampen diagram for uv^{-1} is at distance at most $g-1$ from a vertex in u.

Now we can see how to build a Dehn-reduced submonoid.

Proposition 6.11. *Let $T \subset (S^{\pm})^*$ be such that*

(1) for all $t \in T$, t is Dehn-reduced and has length at least g;
(2) for all $t \in T$, neither its length g prefix nor its length g suffix are a prefix of any word of R_g;
(3) the set of initial letters of words of T is disjoint from the set of final letters.

Then the language T^* is Dehn-reduced. Therefore T^* is weakly-L-proximate, and the submonoid it generates is L-recognisable.

7. Application to rational subsets of right-angled Artin groups

In this section we shall show that in the case of rational subsets of right-angled Artin groups, given an appropriate rational structure L we may construct an algorithm from the procedure described in Section 4.2.

Definition of raag. Description of geodesics in raags. Supp(w) for an element of $(S^{\pm})^*$ and lk(s) for $s \in S^{\pm}$. If w is a word in $(S^{\pm})^*$ then we say that a word w' is obtained from w by a *shuffle reduction* if $w = w_0 a w_1 a^{-1} w_2$ and $w' = w_0 w_1 w_2$, where $a \in S^{\pm}$, $w_i \in (S^{\pm})^*$, for $i = 0, 2$ and $w_1 \in (\mathrm{lk}(a))^*$. In this case we write $w \underset{\mathrm{sr}}{\longrightarrow} w'$. From [**?**] a word $w \in (S^{\pm})^*$ is a geodesic representative of $\pi(w)$ if and only if its length cannot be reduced by a shuffle reduction. A shuffle reduction, with the description above, is said to be *minimal* if w_1 is geodesic. As w_1 may be the empty word a free reduction is a special case of a minimal shuffle reduction. If $w \in (S^{\pm})^*$ then a geodesic representative w' of $\pi(w)$ may be obtained by performing a finite sequence of minimal shuffle reductions to w; that is, there is a sequence $w = w_0, \dots, w_n = w'$, such that $w_i \underset{\mathrm{sr}}{\longrightarrow} w_{i+1}$ is a minimal shuffle reduction, for $0 \le i \le n-1$ and w' is geodesic (see [**?**]). Moreover, the number of (minimal) shuffle reductions required to rewrite w to a geodesic depends only on w and we denote this number by $\mathrm{sr}(w)$. A word $w = aw_1a^{-1}$ where, as above, $a \in S^{\pm}$, $w_1 \in (\mathrm{lk}(a))^*$ and w_1 is geodesic is called a *minimal shuffle reduction word*, and we call its expression as awa^{-1} its *standard factorisation*.

Throughout this section let G be a right-angled Artin group with presentation $\langle S \mid R \rangle$, where R is closed under taking inverses and cyclic permutation of words, let L be the regular language consisting of all geodesics for G and let $\mathcal{A}$ be an automaton over $S^{\pm}$ such that the language Q of $\mathcal{A}$ is L-proximate in G. We do not assume that we know the corresponding constants, k and c, of L-proximacy. The procedure of Section 4.2 produces a sequence $\mathcal{A}_0 = \mathcal{A}, \mathcal{A}_1, \mathcal{A}_2, \dots$ of automata such that, for some $i \ge 0$, the set of all L-representatives of $K = \mu(Q)$ is contained in $L(A_i)$. To make an algorithm of this procedure requires a test to decide whether or not $\mathcal{A}_i$ has the latter property. Our test has two parts, which we call Test 1 and Test 2.

Test 1. Check that $L(\mathcal{A}_i)$ contains at least one L-representative of g, for all $g \in K$.
Test 2. If Test 1 is successful check, for each word w in $L \cap L(\mathcal{A}_i)$, that every element w' of L such that $\mu(w') = \mu(w)$ belongs to $L(\mathcal{A}_i)$.

Clearly $L(\mathcal{A}_i)$ contains all representatives of K if and only if both parts of the test are successful.

To simplify notation assume $\mathcal{A}_i$ is the (deterministic, trim) finite state automaton $(S^{\pm}, V, E, V_F, v_0)$. As in Section 2, we may regard $\mathcal{A}_i$ as a labelled, directed graph (V, E) with labelling function l and we shall need some notation for such graphs. We may refer to the edge $(u, s, v) \in E$ simply as (u, v) when we do not need to consider its label s. For $u \in V$ define the set $\mathrm{out}(u) = \{v \in V \mid (u, v) \in E\}$. As mentioned on page 5 we may identify a path with its image in $\mathcal{A}_i$, which we write as a sequence $u_0, e_1, u_1, \dots, e_n, u_n$, where $u_i \in V$, $e_i \in E$ and $e_i = (u_{i-1}, u_i)$, and as before such a path has length n. We shall sometimes define a path by writing its edge sequence, in the form $p = e_1 e_2 \cdots e_n$, where p is the path above. A path with *initial* vertex $u_0 := x$ and *terminal* vertex $u_n := y$ is called an *(x, y)-path*. If Y is a subset of V then an (x, y)-path such that $y \in Y$ is called an (x, Y)-path. In this terminology $L(\mathcal{A}_i)$ is the set of words w such that w is the label of a (v_0, V_F)-path in $\mathcal{A}_i$.

A path is called *reduced* if its label is reduced as a word in $F(S)$ and *geodesic* if its label is geodesic with respect to the presentation $\langle S \mid R \rangle$. A path is called a *minimal shuffle reduction* path if its label is a minimal shuffle reduction word. Moreover, in this case if $l(p)$ is the minimal shuffle reduction word with standard factorisation awa^{-1} then the factorisation of p as $e_0p_1e_{n+1}$, where $n = |w|$, $e_i \in E$, $l(e_0) = a$, $l(e_{n+1}) = a^{-1}$ and $l(p_1) = w$ is called the *standard factorisation* of p.

SR: reworded

Test 1. Note that we may assume that $i \geq 1$ so that $Q_i := L(\mathcal{A}_i)$ contains all reduced words obtained by free reduction from its elements. To simplify notation set $\mathcal{B} := \mathcal{A}_i = (S^{\pm}, V, E, V_F, v_0)$ in the description of Test 1 that now follows.

Test 1. Let p be an acyclic minimal shuffle reduction (u, v)-path which has standard factorisation $e_0p_1e_{n+1}$, where $n \geq 0$, $e_0 = (u, a, u_0)$, $e_{n+1} = (u_n, a^{-1}, v)$ and p_1 is a (u_0, u_n)-path with label $l(p_1) = w$. We wish to check that, for some vertex v', $\mathcal{B}$ has a (u, v')-path q_w with label w and an edge $e'_{n+1} = (v', a, u_n)$. If this is true then, as $\mathcal{B}$ is the result of an application of Benois' theorem and the path $e'_{n+1}e_{n+1}$ has label aa^{-1}, it follows that for each edge (v, s, v'') of $\mathcal{B}$ there is a corresponding edge (v', s, v'').

We shall perform our verification for all geodesic (u_0, u_n)-paths with labels in $\mathrm{lk}(a)^*$. We note that the set of words which can appear as labels of such paths is regular, being the set of geodesic elements accepted by the automaton $\mathcal{X} = (lk(a), V, E_a, \{u_n\}, u_0)$, where E_a is the set of edges of $\mathcal{B}$ with labels in $\mathrm{lk}(a)$. For our test, we want to show that for every geodesic word z of $L(\mathcal{X})$ there is a (u, U_a)-path in $\mathcal{B}$ with label z, where U_a is the set of vertices $\{x \in V \mid \exists (x, a, u_n) \in E\}$. But the set of geodesic (u, U_a)-paths is also regular, being the language of geodesic words accepted by the automaton $\mathcal{Y} = (\mathrm{lk}(a), V, E_a, U_a, u)$. It will be enough, then, to check that $L \cap L(\mathcal{X})$ is contained in $L \cap L(\mathcal{Y})$. There are standard methods for constructing the intersection or regular langauges and for testing whether one regular subset of $(S^{\pm})^*$ is contained in another (ref Hopcroft & Ullman), so this check can be performed.

Our test, then, proceeds as follows. For each $a \in S^{\pm}$ and all $u, v, u_0, u_n \in V$, if there exist edges $e_l = (u, a, u_0)$ and $e_r = (u_n, a^{-1}, v)$ then check that $L \cap L(\mathcal{X}) \subseteq L \cap L(\mathcal{Y})$, where $\mathcal{X}$ and $\mathcal{Y}$ are as above. If the test is passed for all a and all such pairs of edges e_l and e_r then the verification is complete and $\mathcal{B}$ passes Test 1.

AD: Test 1 is too strong. The original idea was to test that all geodesics obtained from words in $L(\mathcal{A}_0)$ by sequences of minimal shuffle reductions belong to $\mathcal{B}$. For that the test should only be on minimal shuffle reduction paths which are involved in such sequences of reductions. As it is it's applied to all minimal shuffle reduction paths.

To prove that this test succeeds if and only if $\mathcal{B}$ contains at least one L-representative of every element of K we need to demonstrate that (i) it fails if there is an element $g \in K$ such that $\mu^{-1}(g) \cap L(\mathcal{B})$ contains no element of L and (ii) it succeeds if $\mu^{-1}(g) \cap L(\mathcal{B}) \cap L \neq \emptyset$, for all $g \in K$.

AD: got to here

Test 2. Check there are no missing squares (after intersecting with L).

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